



MINING DEALER BEST PRACTICE

Mounting Fixture for Metal Spray Gun

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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October 2023
4.02-231002-2

1.0 Introduction

This Best Practice document contains information on how to build a mounting fixture (as shown here for a C175 engine block - Figure 1 and 9) to affix a metal spray gun to an industrial robot.

The use of industrial robot for metal spray purposes allows for precise control and accurate material velocity to create a strong bond coat which is evenly distributed.

2.0 Best Practice Description

Use the fixture to locate the metal spray gun in a way that the robot can utilize all available axes keeping the correct orientation of the spray gun to complete the spray coat.

The manufacture process to build the fixture consists of cutting a 0.5-inch aluminum plate to create an assembly shown as follows:



Figure 1. Mounting Fixture Assembly

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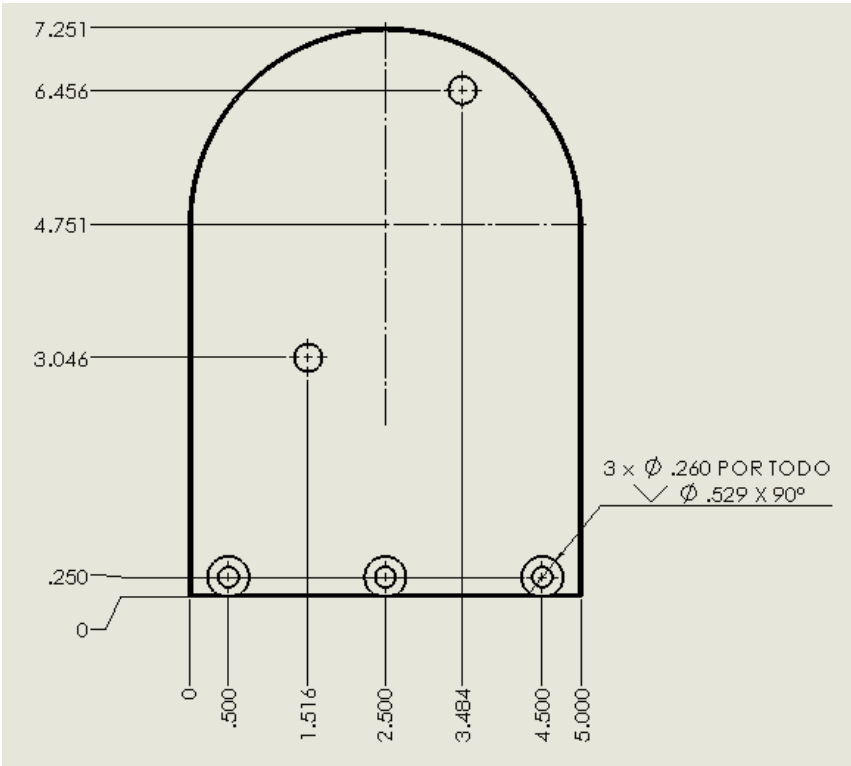


Figure 2 - Units are Inches

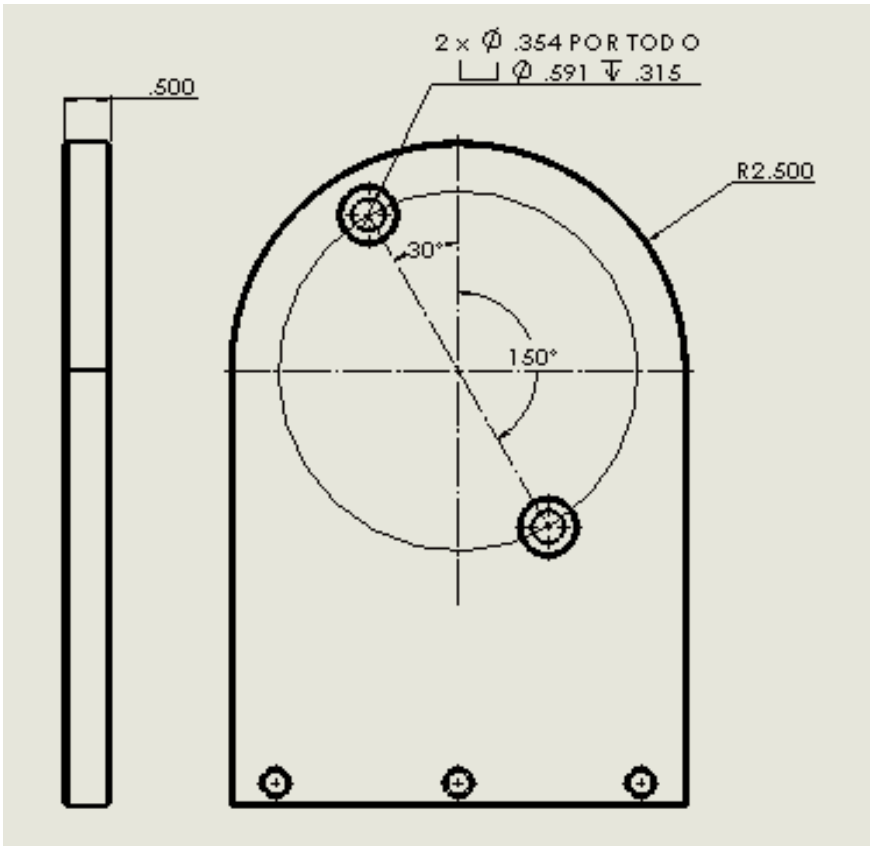


Figure 3 - Units are Inches

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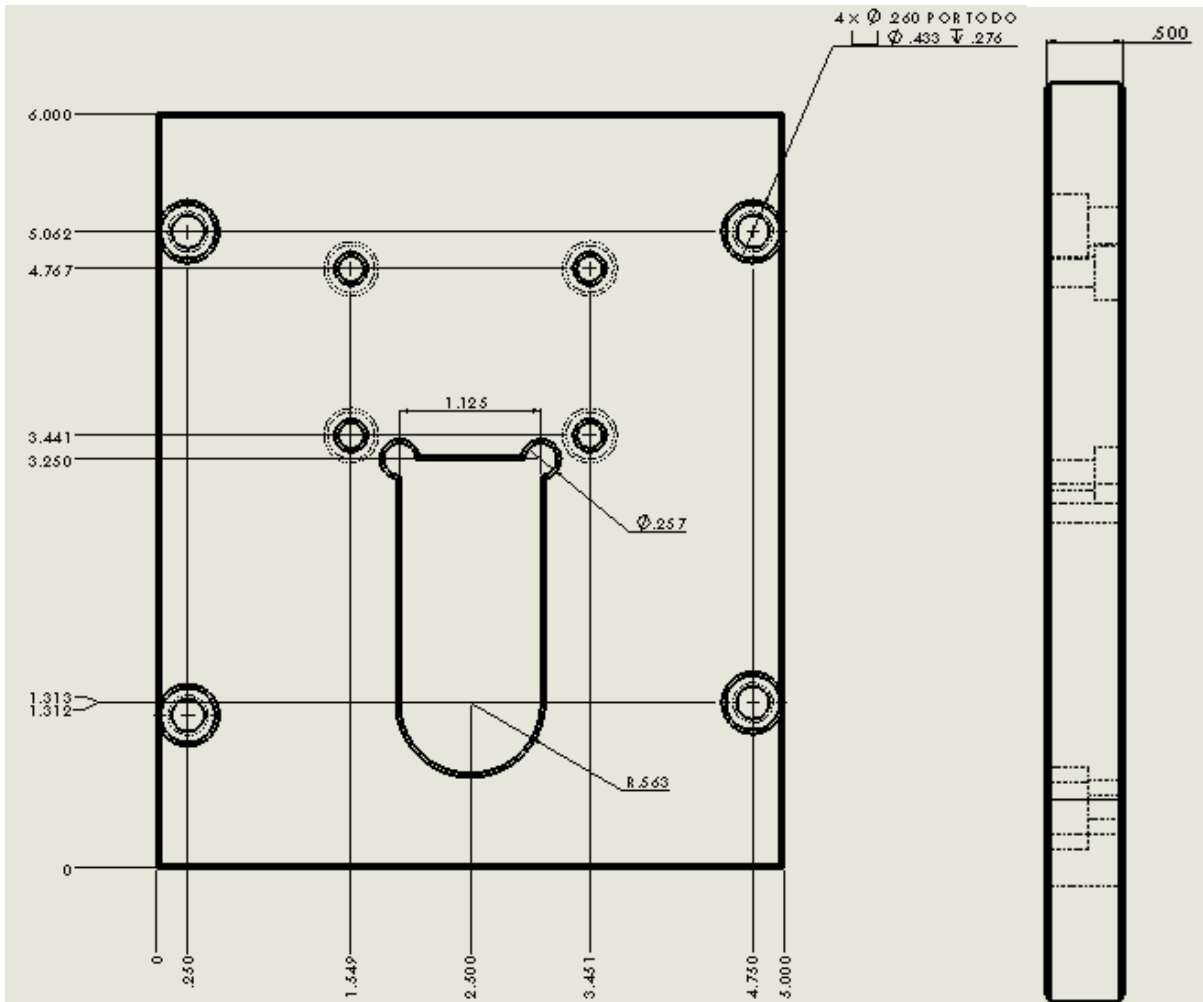


Figure 5 - Units are Inches

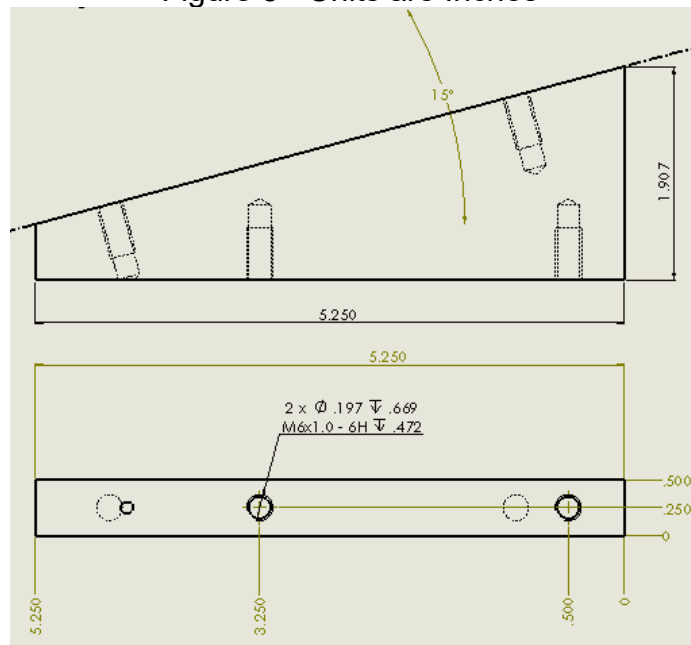


Figure 6 - Units are Inches

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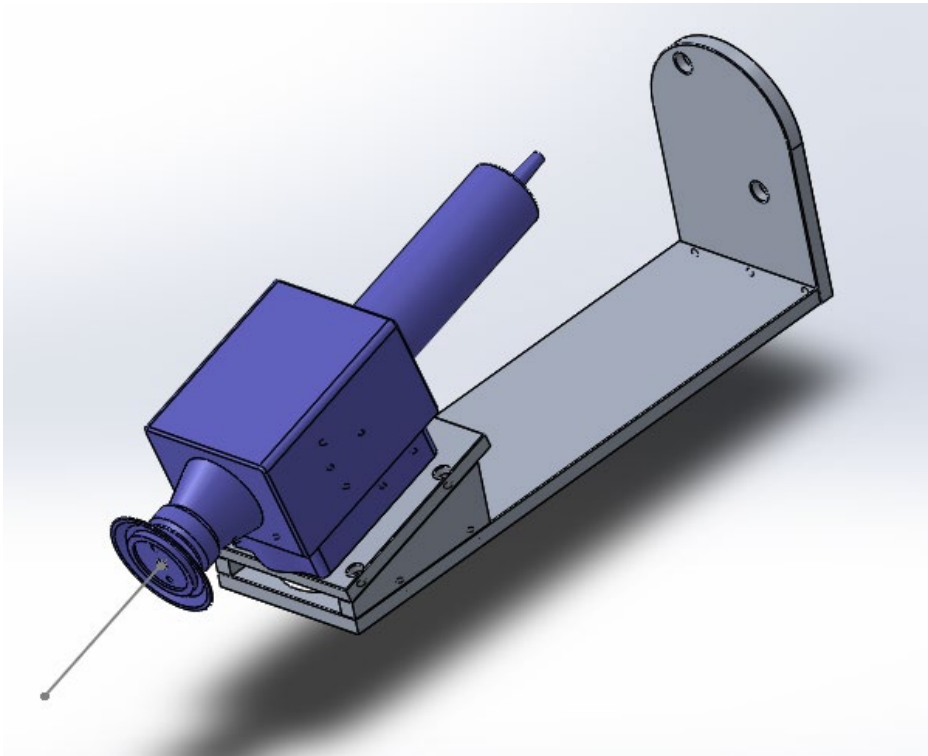


Figure 7

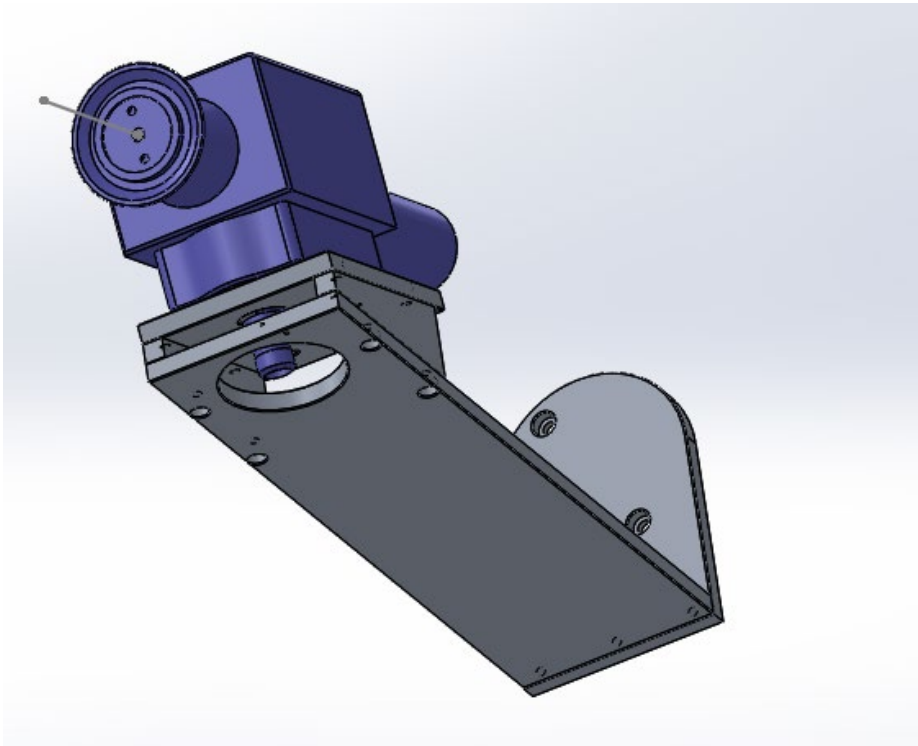


Figure 8

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3.0 Implementation Steps

Use 0.5-inch aluminum plate and machine the different pieces of the assembly.

Drill and tap the holes.

Assemble the mounting fixture by welding all pieces together.

Clamp the fixture into the 6th axis of the robot, using mounting screws.

Assemble the Smart Arc Gun onto the fixture as shown.



Figure 9. Smart Arc Gun mounted on the fixture application in block Spot Face

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4.0 Benefits

- Increases the reach capabilities of the robot
- Allows spraying of the cylinder block liner bores of C175 engines
- Improves the metal spray Set-Up times, improvement by 30%

5.0 Resources Required

Aluminum plate (0.5-inch thickness)
 Milling machine
 Metal spray equipment
 Industrial robot
 Welder with aluminum welding capabilities
 Manufacturing Price ~ 650 USD

6.0 Supporting Attachments / References

- SEBF9238 "Fundamentals of Arc Spray for Reconditioning Components" [SIS 2.0 \(cat.com\)](#)
- Drawings see file attachment - Units are Inches

7.0 Related Best Practices

4.02-231002-1 Setting Up the Metal Spray Robot Using a Cone Tool

8.0 Acknowledgements

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